

ENTERPRISE OPERATIONS MANUAL

Manufacturing Expense & Cost Control ERP

Complete Architecture, Workflow Lifecycles, and Module Operations with Live Screenshots

SYSTEM

MECS Platform v1.0

DEPLOYMENT

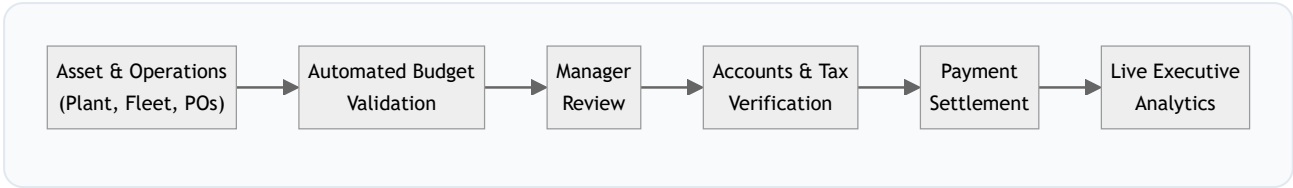
Production / Cloud ERP

GENERATED

September 2026

1 End-to-End Operational Flow & Architecture

The Manufacturing Expense & Cost Control ERP connects all operational departments—Production, Maintenance, Fleet, Warehouse, and Administration—into a consolidated financial approval and cost control pipeline.



Operational Lifecycle Stages

1. Initiation & Asset Logging

Operational claims, machine spares usage, fuel fills, and purchase orders are logged at source with vendor invoices.

2. Real-Time Budget Check

System automatically verifies department budget availability before submission proceeds to the approval hierarchy.

3. Multi-Tier Approval Chain

Claims pass through Manager verification, Accounts tax validation, and Super Admin review for high-value transactions.

4. Settlement & Analytics

Cleared payments are recorded (NEFT/Cheque), immediately reflecting in budget burn rates and spend anomaly engines.

2 Executive Dashboard & Spend Analytics

The executive dashboard provides unified visibility into overall company expenditures, month-over-month comparisons, budget consumption, and rolling anomaly detections.

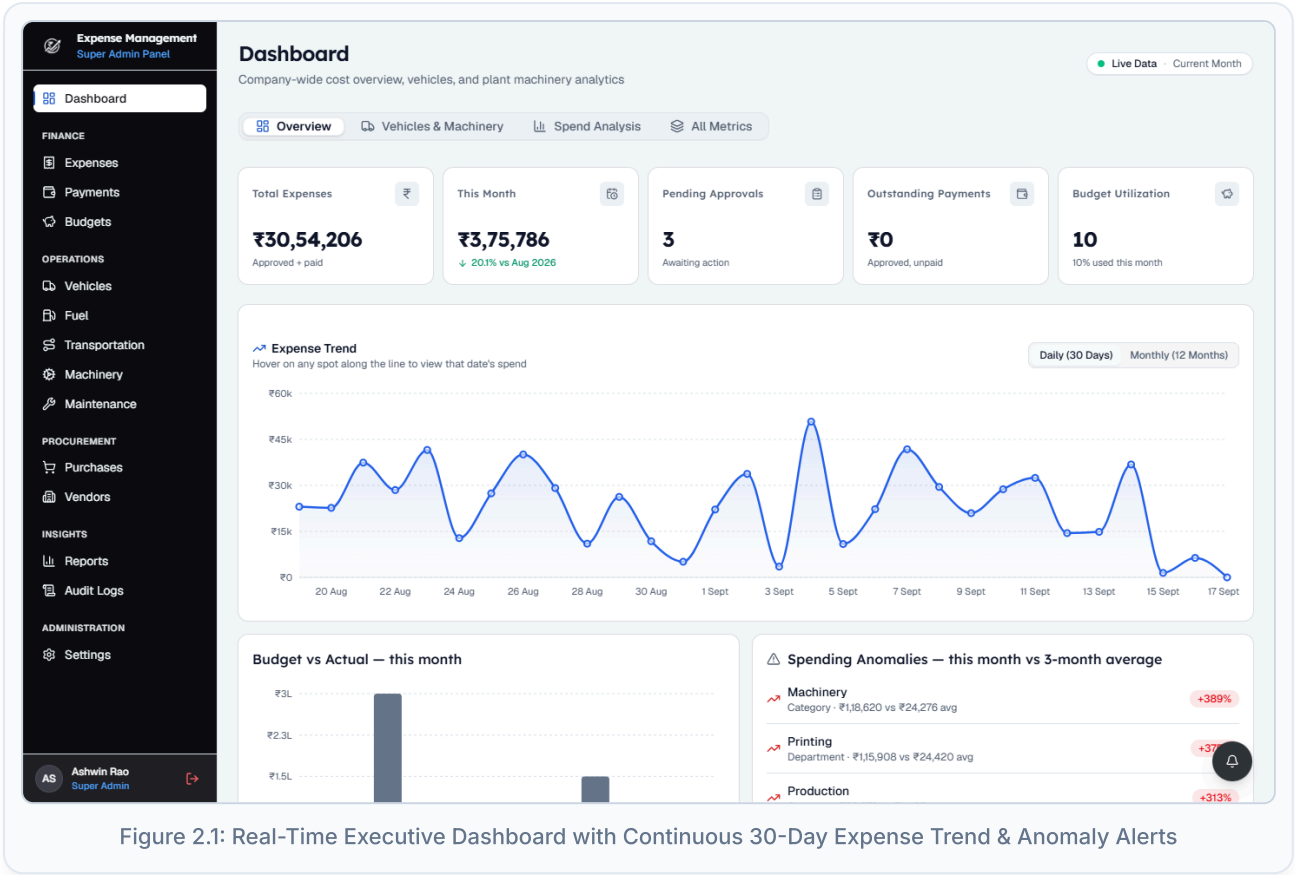


Figure 2.1: Real-Time Executive Dashboard with Continuous 30-Day Expense Trend & Anomaly Alerts

Key Capabilities

- Executive KPI Cards:** Live tracking of Total Expenses (₹30,54,206), Current Month Spend (₹3,75,786 with explicit ↓ 20.1% vs Aug comparison), Pending Approvals, and Budget Utilization.
- Interactive 30-Day Expense Trend:** High-resolution continuous graph lines with spot hover tooltips displaying the exact expenditure for any specific date.
- Automated Anomaly Detection:** Automatically flags category and department expenditures that surge abnormally against a rolling 3-month baseline (e.g., Machinery +389%, Printing +37%).
- Ergonomic Sidebar:** Clean navigation with a full-bleed charcoal profile rectangle, crisp white typography, and a dedicated red door logout icon.

3 Expense Management & Multi-Tier Approvals

The expenses module handles company-wide cost claims with document uploads, automated status progression, and priority visual highlights.

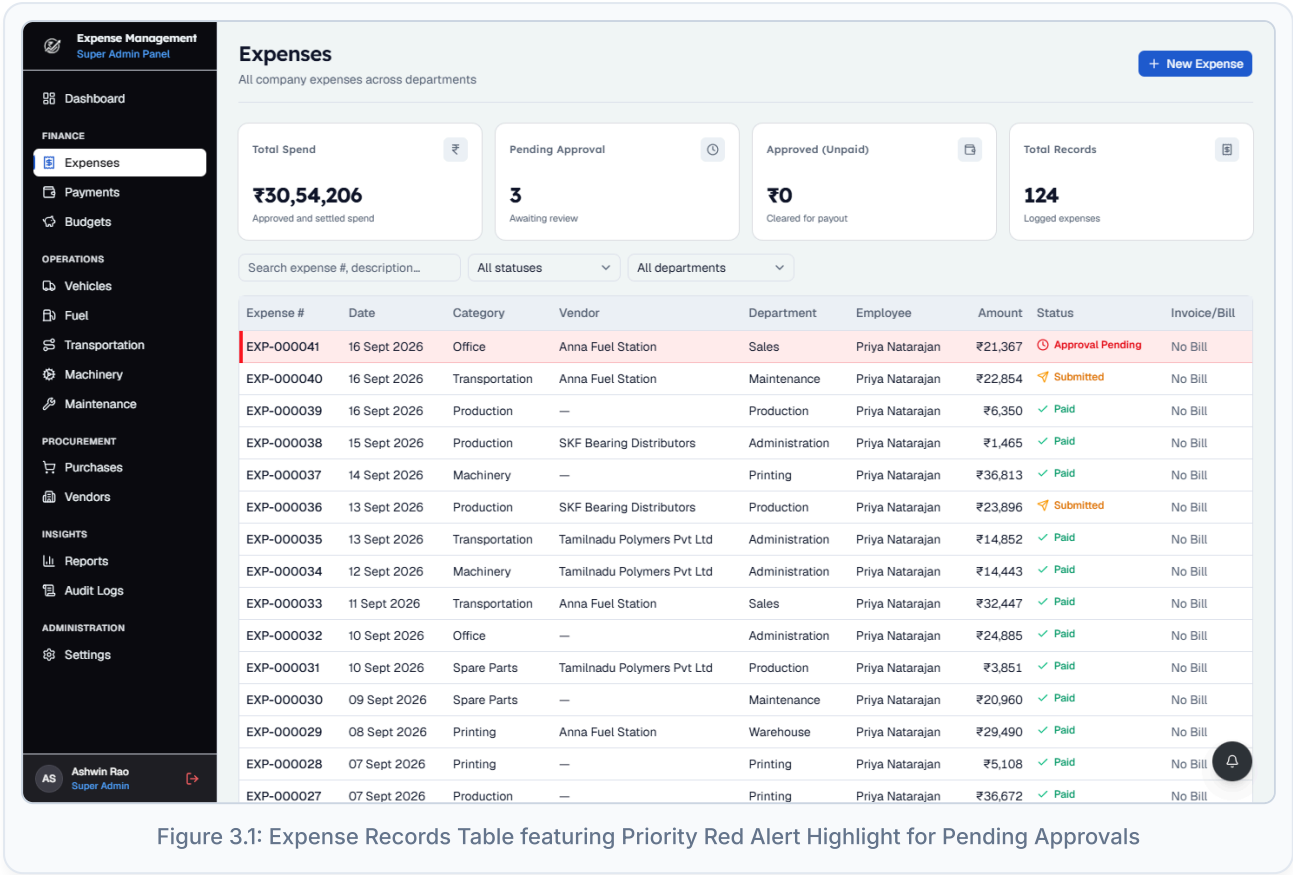


Figure 3.1: Expense Records Table featuring Priority Red Alert Highlight for Pending Approvals

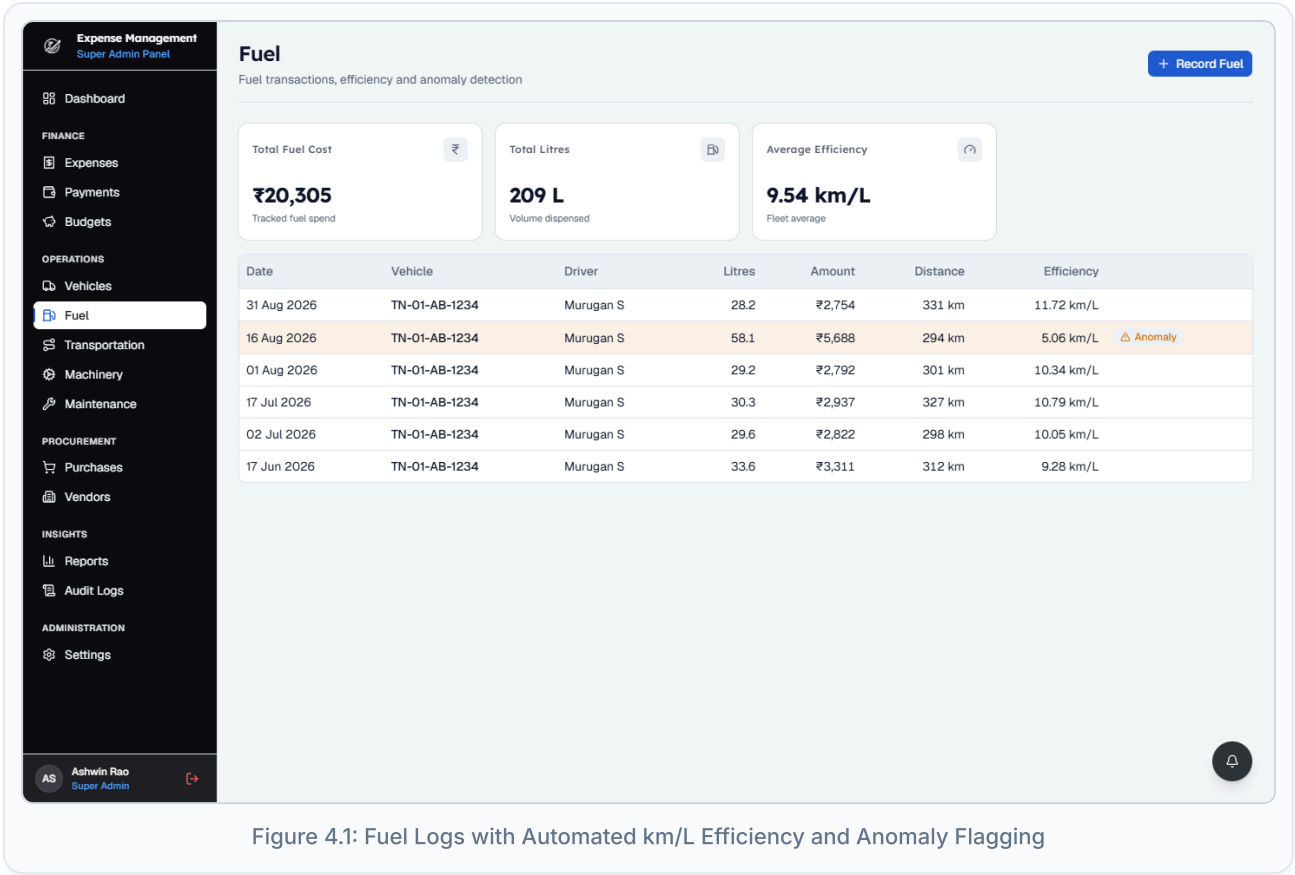
Workflow & Governance

- **Dynamic Multi-Filter:** Search by expense number, vendor, department, or date range with real-time responsive updates.
- **Pulsing Alert for Pending Review:** Records in Approval Pending state feature a high-priority red perimeter highlight to accelerate clearance.
- **Lifecycle Progression:** Submitted → Approval Pending → Approved (Unpaid) → Paid.
- **Compliance & Bill Proof:** Direct integration with cloud document storage for receipts, delivery challans, and GST bills.

4

Fleet Logistics & Fuel Efficiency Engine

Monitors company transport vehicles, fuel dispensations, driver efficiency, and statutory document renewals to prevent pilferage and regulatory penalties.

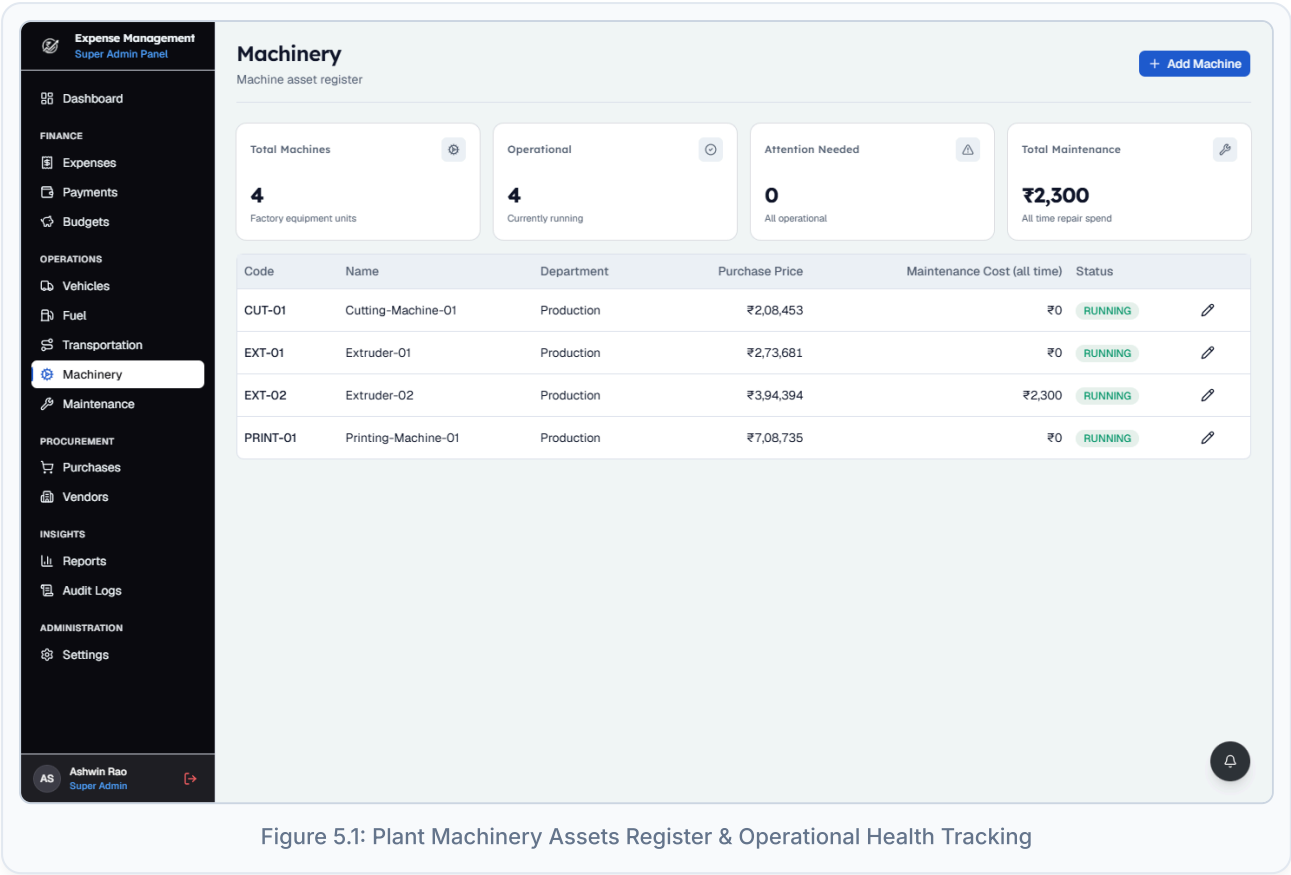


Workings & Algorithms

- Automated Efficiency Calculation:** Divides trip distance covered by fuel volume dispensed to calculate exact km/L per log.
- Fuel Pilferage & Fault Detection:** Entries dropping significantly below the vehicle's historical baseline (e.g. 5.06 km/L vs 11.72 km/L fleet benchmark) trigger an immediate Anomaly alert.
- Compliance Monitoring:** Proactively tracks *Pollution certificate expiry*, *Fitness certificate expiry*, and insurance renewals.

5 Plant Machinery Assets & Operations

Maintains a centralized registry of manufacturing machinery, operational status tracking, installation dates, and department allocations.



Workings & Asset Governance

- **Asset Registry:** Tracks machine name, internal asset code (e.g. EXT-01, CUT-01), department assignment, purchase price, and installation history.
- **Live State Monitoring:** Operational status tags (Running, Maintenance, Idle) keep plant managers informed of line availability in real time.
- **Asset Cost Linking:** Automatically aggregates cumulative maintenance and parts costs against each individual machine asset over its lifetime.

6 Preventative & Breakdown Maintenance

Automates work order ticketing, servicing schedules, spare parts inventory deduction, and external labour cost tracking.

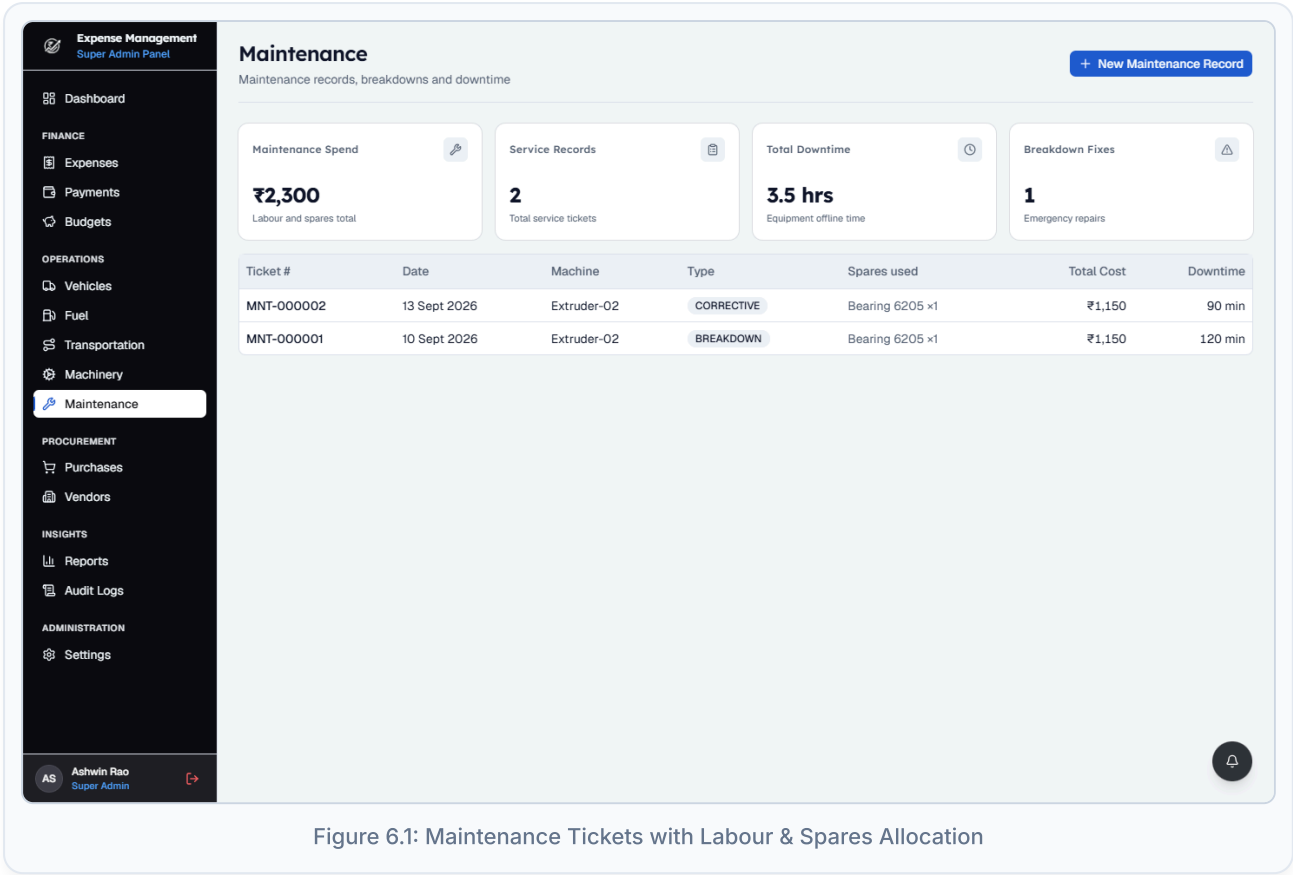


Figure 6.1: Maintenance Tickets with Labour & Spares Allocation

Workings & Cost Tracking

- **Preventative Maintenance (PM):** Scheduled routine servicing to prevent costly line breakdowns and manufacturing downtime.
- **Breakdown & Corrective Work Orders:** Rapid ticketing for unexpected faults with tracked downtime hours and priority response flags.
- **Total Cost Reconciliation:** Captures spare parts consumed (bearings, seals, heating elements) alongside technician service fees, automatically posting costs to the plant operating ledger.

7 Procurement & Purchase Order Flow

Automates vendor engagement from purchase request initiation through goods receipt notes (GRN) and invoice three-way matching.

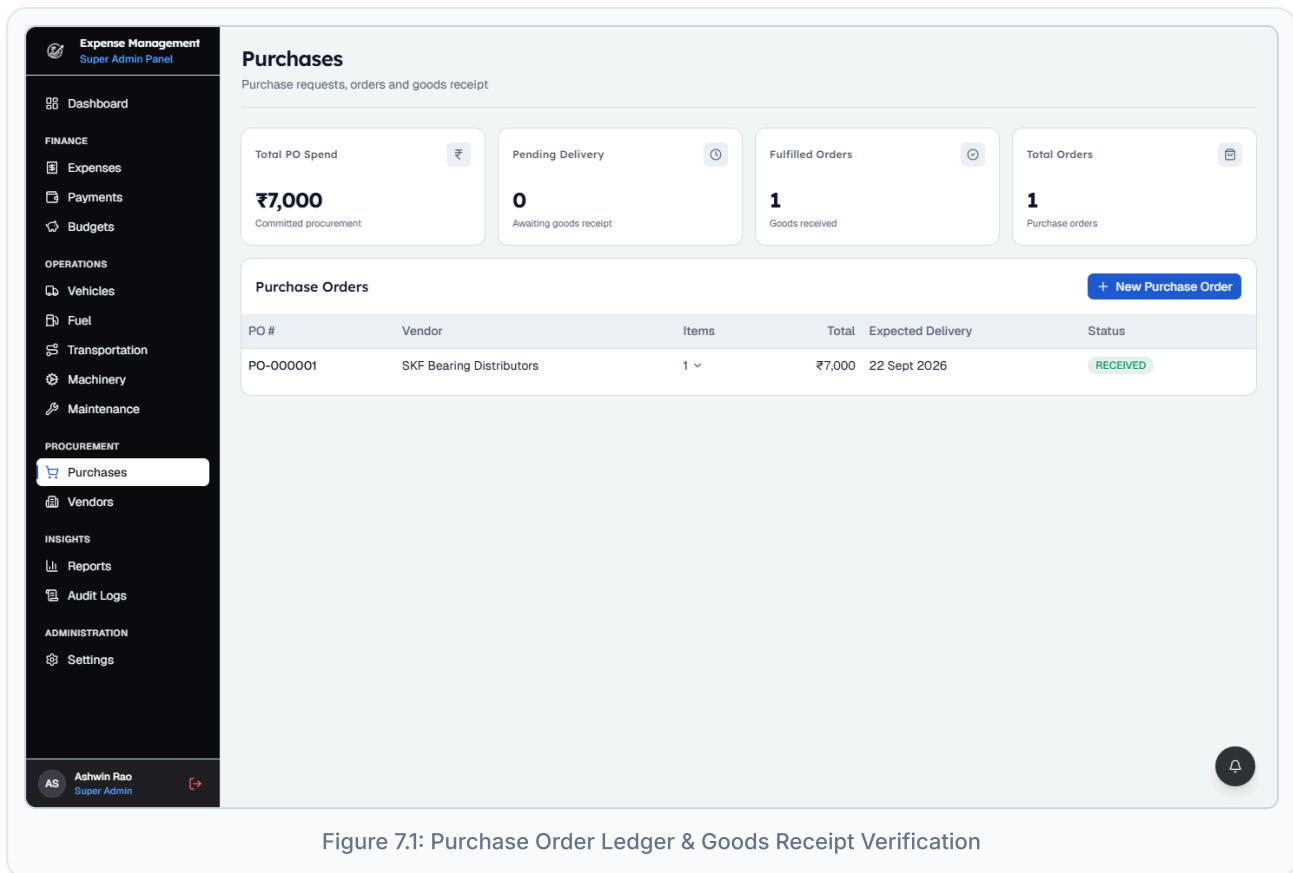


Figure 7.1: Purchase Order Ledger & Goods Receipt Verification

Workings

- **PO Issuance:** Formal PO creation with tax calculations (CGST/SGST/IGST), vendor details, delivery schedules, and payment terms.
- **Goods Receipt Note (GRN):** Warehouse verifies physically delivered items against PO quantities before accounts clearance.
- **Three-Way Match:** System matches PO → Goods Receipt → Vendor Tax Invoice before payments can be disbursed.

8 Budgets & Role-Based Access Control (RBAC)

Financial safeguards and organizational access boundaries enforced at database and route levels.

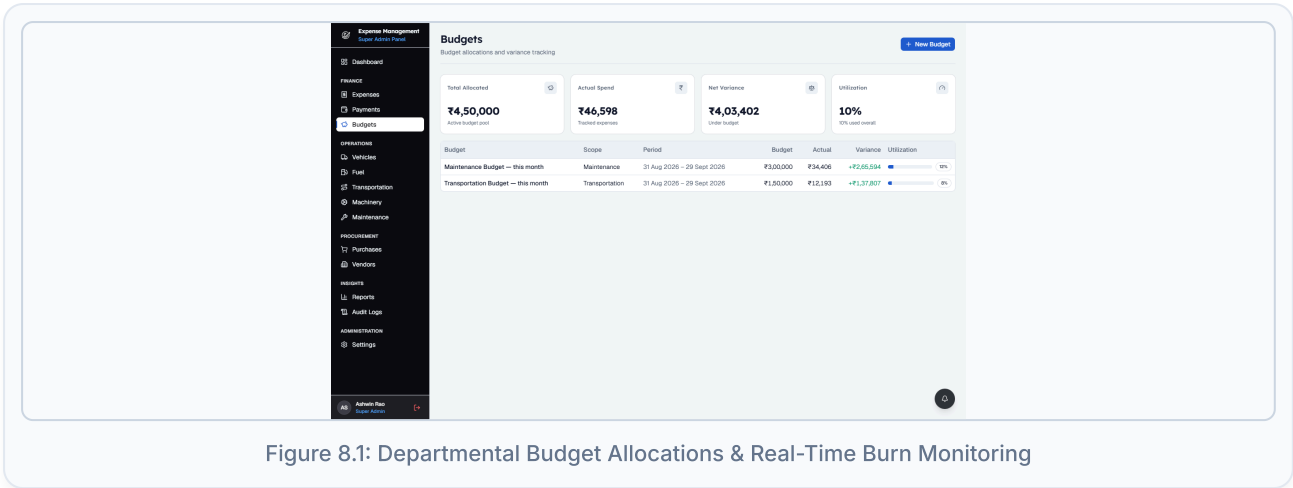


Figure 8.1: Departmental Budget Allocations & Real-Time Burn Monitoring

Role Permissions Matrix

Role	Expenses	Fleet & Fuel	Machinery	Purchases	Budgets
Super Admin	Full Control	Full Control	Full Control	Full Control	Full Control
Admin	Approve All	Full Control	Full Control	Full Control	Allocate
Accounts	Verify & Pay	Audit & Pay	Spares Audit	Three-Way Match	Monitor
Purchase Manager	Dept Submit	View	Parts PO	Full POs	PO Budget
Maintenance Manager	Dept Submit	View	Work Orders	Requisitions	Dept Budget
Transport Manager	Dept Submit	Fleet Ops	View	Transport PO	Dept Budget
Employee	Personal Submit	Personal Log	View	—	—